

Boundary-Inversion Cosmology VI: A Sqrt-Reg EFT-Clock Inflationary Branch and Calibration-Frame Late-Universe Extension

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Abstract

We present Boundary-Inversion Cosmology VI (BIH-VI) as a boundary-anchored effective-field-theory cosmology built around the invariant marginal-horizon seam $\chi = h^{\{ab\}} \nabla_a R \nabla_b R = 0$, equivalently $R = 2G E_{MS}$. The early-universe branch realizes a Sqrt-Reg entropy-selected density shield through a stable EFT clock with $H^2 = H_c^2 y(2-y)$, $y_N = -m(y)y$, and $\epsilon_H = m(1-y)/(2-y)$. For the frozen benchmark $m(y)=\mu(1-y)^3$, $\mu=1$, $c_s=1$, and $N_*=55$, the model gives $n_s \approx 0.9652$, $r \approx 1.4e-3$, and $\alpha_s \approx -5.8e-4$. The late-universe branch is formulated only as a falsifiable calibration-frame extension: the universal metric-lapse route is rejected at the diagnostic level, while C3 isolates $\Delta M_B \approx 0.18$ mag as a local readout target and uses $Q_{bulk}(z)$ smoothness as the thermodynamic falsifier.

Scope boundary. BIH-VI is presented as an EFT-level cosmological framework. It does not claim to be a UV-complete microscopic theory. The early-universe branch is frozen at the Sqrt-Reg/EFT-clock benchmark. The late-universe branch is a diagnostic calibration-frame extension motivated by the failure of universal metric-lapse deformations. The calibration-frame branch is falsifiable through full distance-likelihood tests and through the smoothness, boundedness, and physical interpretability of $Q_{bulk}(z)$.

1. Introduction

The paper is organized as a hierarchy of claims: frozen early branch, late-universe stress test, metric-lapse rejection, calibration-frame extension, and Q_{bulk} falsifiability. This draft intentionally avoids claiming a solved Hubble tension or a UV-complete substrate theory.

Pivot paragraph: Having established the stable EFT-clock realization of the Sqrt-Reg density shield as an early-universe branch, we now assess whether the same boundary-centered framework can be extended to late-time cosmology without modifying its frozen inflationary mechanics. The low-density expansion of the Sqrt-Reg background shows that the entropy shield acts as a UV regulator rather than an infrared dark-energy mechanism. We therefore treat the late universe as a separate observational gate on the BIH clock/stress sector. A direct metric-lapse interpretation, in which a redshift-dependent clock factor multiplies all distance integrals, is rejected at the diagnostic level by supernova-shape and BAO-geometry constraints. Consequently, we define a calibration-frame

branch: the global geometry remains governed by $H_{\text{geom}}(z)$, while the local distance ladder is allowed to measure a substrate-dependent calibration offset $\Delta M_B \sim 0.18$ mag. This recasts the Hubble tension as a testable calibration-frame hypothesis rather than a universal modification of light-propagation distances.

2. Boundary seam and Sqrt-Reg thermodynamics

$$\chi = h^{\{ab\}} \nabla_a R \nabla_b R$$

$$\chi = 1 - 2G E_{\text{MS}}/R$$

$$\chi = 0 \Leftrightarrow R = 2G E_{\text{MS}}$$

$$\theta_+ \theta_- = -4 \chi/R^2 \quad [\text{for } l_+ \cdot l_- = -2]$$

The real invariant seam is $\chi=0$. The B-map of null-expansion signs is useful as orientation language, but it is not itself the invariant physical seam.

$$S_{\text{sqrt}}(A) = (1/(4G))[\text{sqrt}(A(A-A_c)) + A_c \text{arcosh} \text{sqrt}(A/A_c)]$$

$$S_A = 1/[4G \text{sqrt}(1-A_c/A)], \quad G_{\text{eff}} = 1/(4S_A)$$

$$F_{\text{sqrt}}(x) = 2 x_c [1 - \text{sqrt}(1 - x/x_c)]$$

$$H^2 = (8 \pi G/3) \rho [1 - 2 \pi G \rho/(3 H_c^2)]$$

3. Frozen EFT-clock branch

$$S_{\text{EFT}} = \int d^4x \text{sqrt}(-g)[M_{\text{Pl}}^2 R/2 - \Lambda(t) - c(t)g^{\{00\}} + (M^2/2)(\delta g^{\{00\}})^2 + \dots]$$

$$H^2 = H_c^2 y(2-y), \quad y_N = -m(y)y, \quad \epsilon_H = m(1-y)/(2-y)$$

$$m(y) = \mu(1-y)^p, \quad \text{benchmark: } p=3, \mu=1, c_s=1, N_*=55$$

$$n_s = 0.9652, \quad r = 1.4\text{e-}3, \quad \alpha_s = -5.8\text{e-}4$$

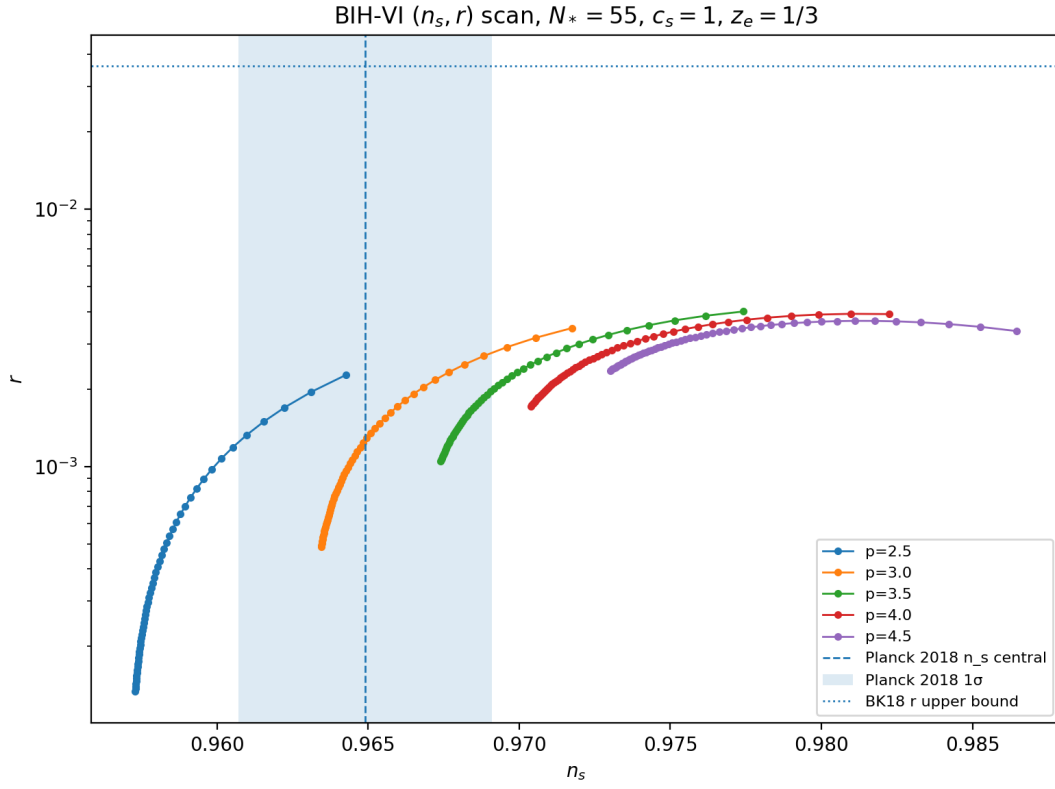


Figure. Frozen early-branch parameter island in the (n_s, r) plane.

4. Exit switch and UHF relic

$$m(y) = \mu(1-y)^3 + m_{\text{post}}/[1 + \exp((y-y_g)/\Delta)]$$

Benchmark: $y_g = 2/3$, $\Delta = 0.01$, $m_{\text{post}} = 4$

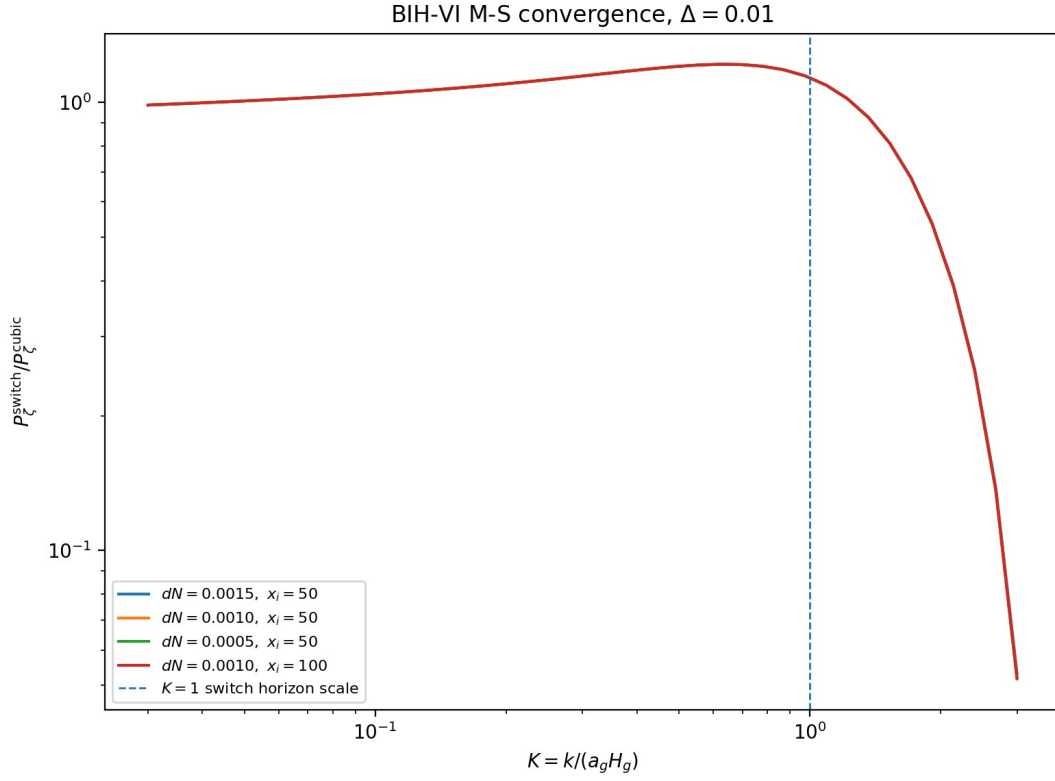


Figure. Mukhanov-Sasaki convergence for the exit switch.

$S_3^{\text{switch}} \supset (M_{\text{Pl}}^2/2) \int dt d^3x a^3 \epsilon_{\text{NL}} \dot{\eta}_H \zeta^2 \dot{\zeta}$

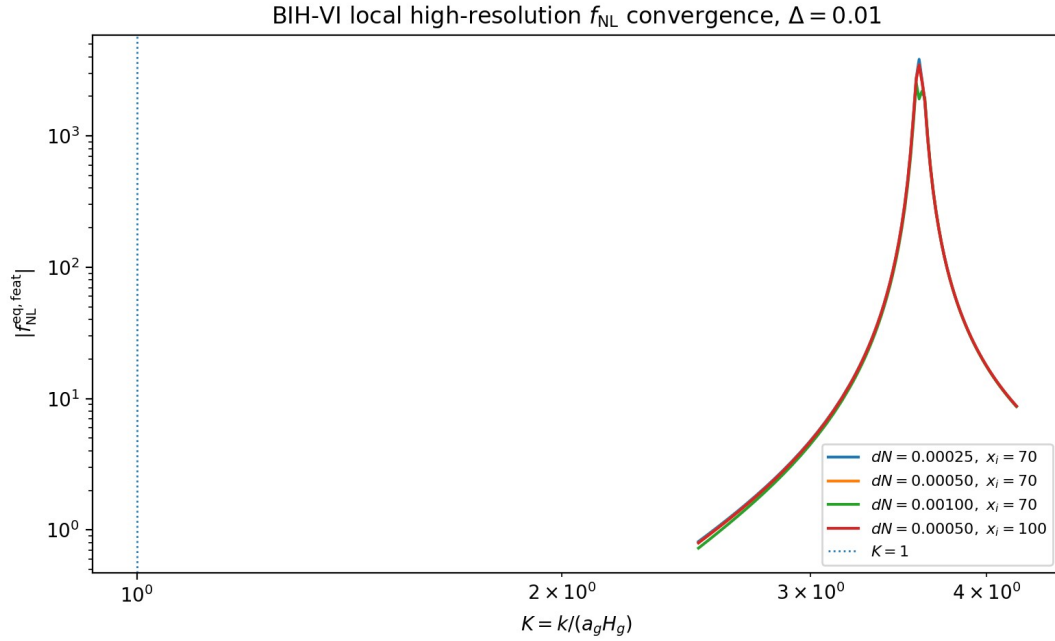


Figure. High-resolution exit feature-bispectrum diagnostic.

$k_{\text{peak}} \sim 1.3e23 \text{ Mpc}^{-1}$, $f_{\text{peak}} \sim 2.0e8 \text{ Hz} \sim 200 \text{ MHz}$

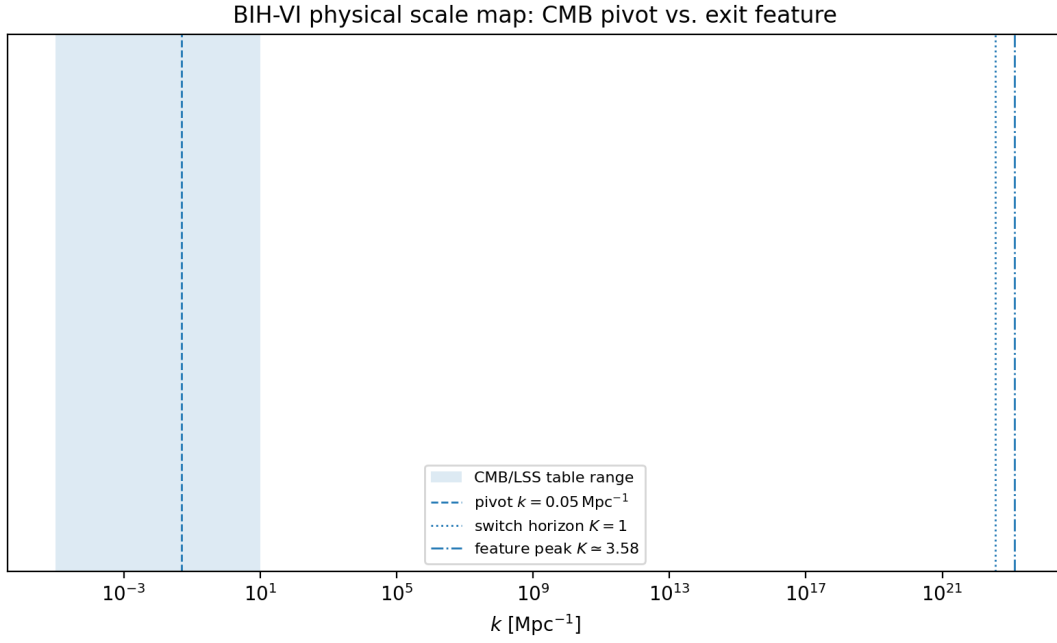


Figure. Scale mapping for the exit feature.

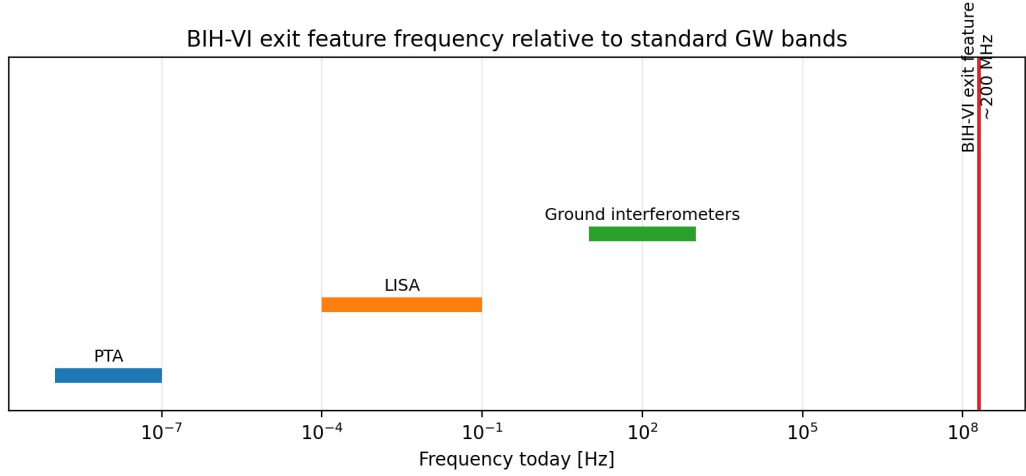


Figure. Present-day UHF frequency mapping for the exit relic.

5. Late-universe diagnostic gate

The Sqrt-Reg shield fades in the infrared and does not replace Lambda by itself. The late-universe sector is therefore a separate observational gate, not a retuning of the frozen early branch.

5.1 Metric-lapse rejection

$H_{\text{obs}}(z) = \nu(z) H_{\text{geom}}(z)$ [rejected as universal distance deformation]

$D_C(z) = c \int_0^z \frac{dz'}{H_{\text{geom}}(z')}$

Diagnostic scans show the required local uplift generates either SN-shape strain or BAO-geometry strain. This branch should remain in the paper as a negative result.

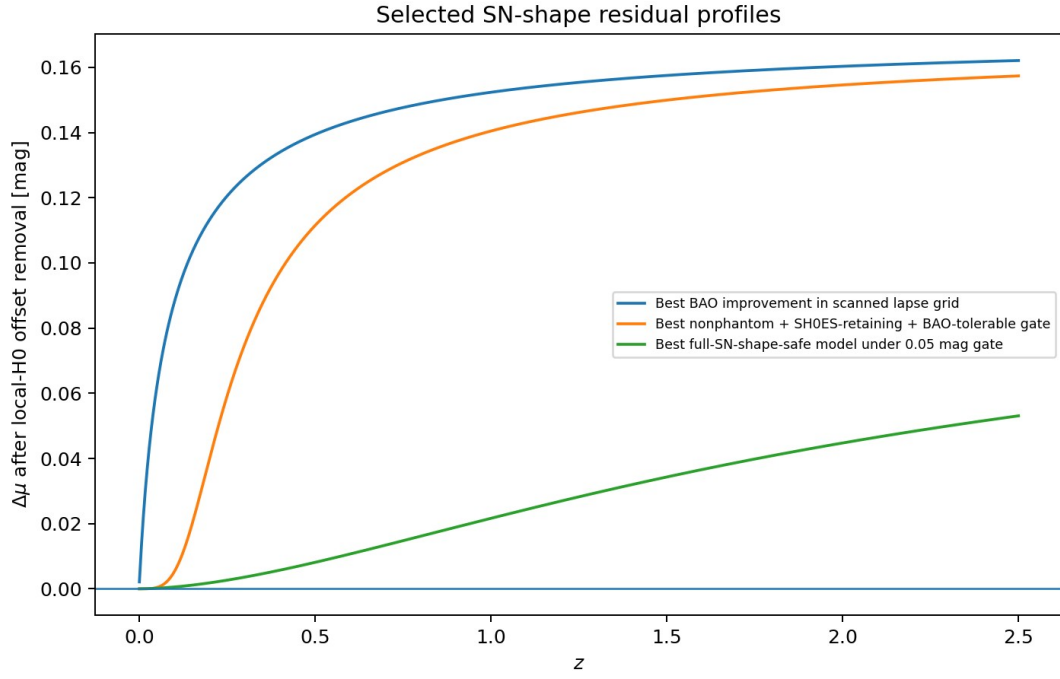


Figure. SN-shape residual diagnostic for the metric-lapse branch.

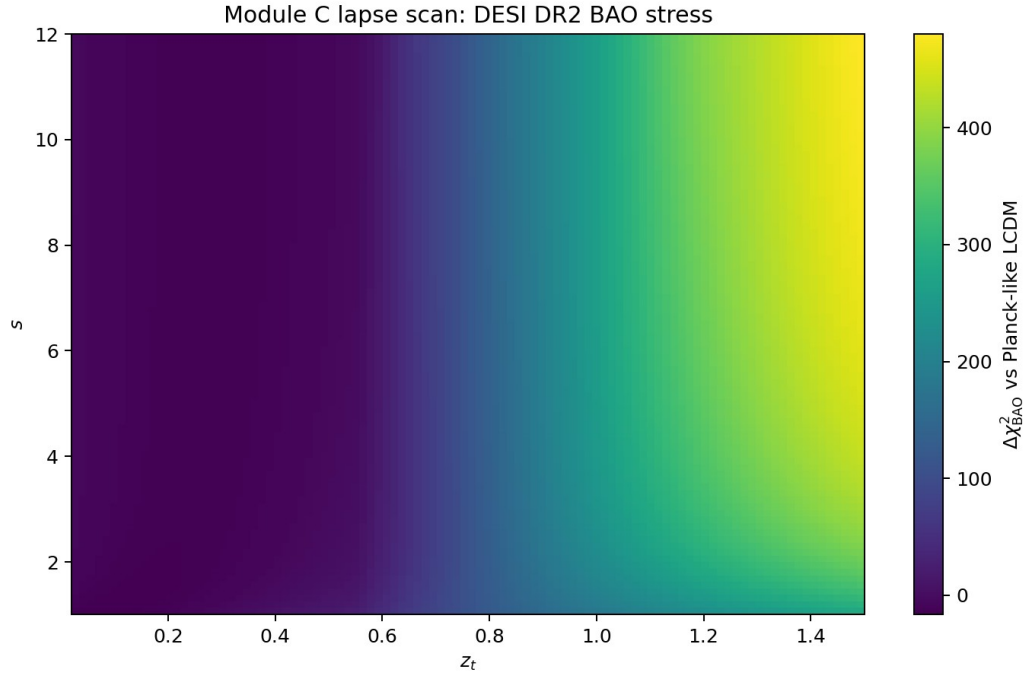


Figure. BAO geometry stress for the metric-lapse branch.

6. C3 calibration-frame extension

$\text{nu_metric}(z) = 1$

$H_0_\text{ladder} = \text{nu_cal } H_0_\text{geom}$

$\text{Delta } M_B = 5 \log_{10}(\text{nu_cal})$

$$\text{nu_cal} = 73.17/67.36 = 1.08625, \quad \text{Delta M_B} = 0.1797 \text{ mag}$$

Delta M_B is a calibration-frame target, not yet a first-principles prediction. It must be tested against independent anchors, host-environment systematics, standard sirens, Cepheid metallicity, TRGB, and masers.

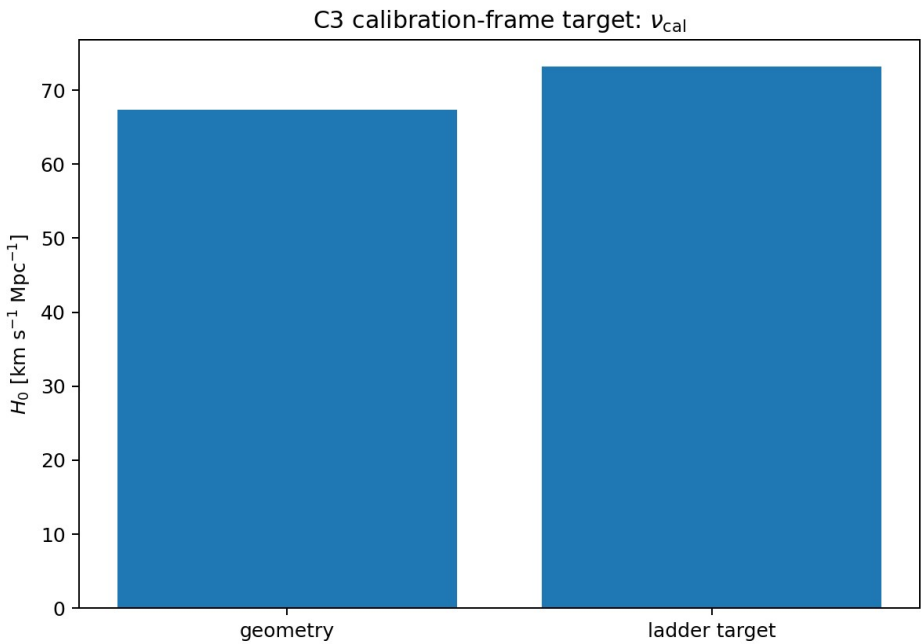


Figure. Calibration-frame target $H0_{\text{ladder}} = \text{nu_cal } H0_{\text{geom}}$.

$$w_{\text{eff}}(z) = w_0 + w_a \, z/(1+z)$$

$$E_{\text{BIH}}^2(z) = \Omega_{\text{r}}(1+z)^4 + \Omega_{\text{m}}(1+z)^3 + \Omega_{\text{k}}(1+z)^2 + \Omega_{\text{DE},0} \, f_{\text{DE}}(z)$$

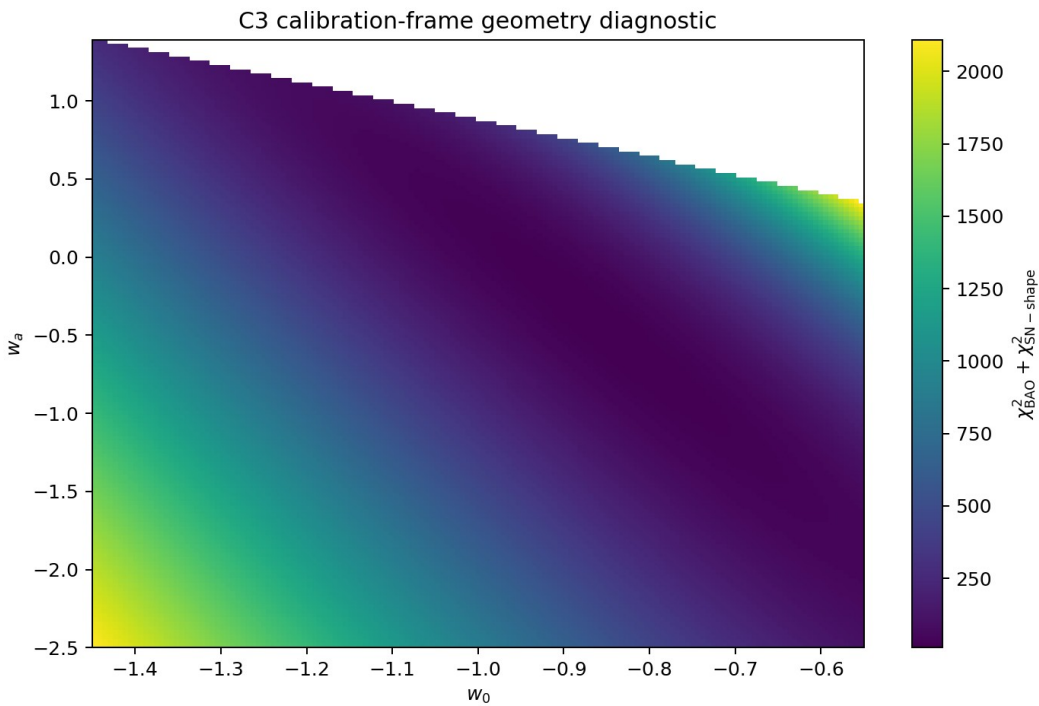


Figure. C3 geometry diagnostic in the CPL scaffold.

7. Likelihood architecture

Theta_C3 = {h, Omega_m, Omega_k, omega_b, w0, wa, M_B^geom, Delta M_B}

Production likelihood must include full Pantheon+SH0ES covariance, DESI DR2 BAO, CMB acoustic-scale priors, and independent calibration-anchor priors. Diagnostic scans should be labeled diagnostic until production chains exist.

Sector	Parameters	Role
Geometry	h, Omega_m, Omega_k, omega_b, w0, wa	BAO/SN/CMB distances
Calibration	M_B^geom, Delta M_B	local readout offset
Falsifier	Q_bulk(z)	thermodynamic consistency

8. Q_bulk reconstruction

$\dot{\rho}_{DE} + 3H(1+w_{int})\rho_{DE} = Q_{bulk}$

$Q_{bulk}/(H \rho_{DE}) = 3(w_{int} - w_{eff})$

For $w_{int} = -1$: $Q_{bulk}/(H \rho_{DE}) = -3[1+w_{eff}(z)]$

C3 survives only if the posterior-implied exchange history is smooth, bounded, low-frequency, and physically interpretable. Jagged, discontinuous, or high-frequency sign-flip chatter falsifies the substrate-stress interpretation.

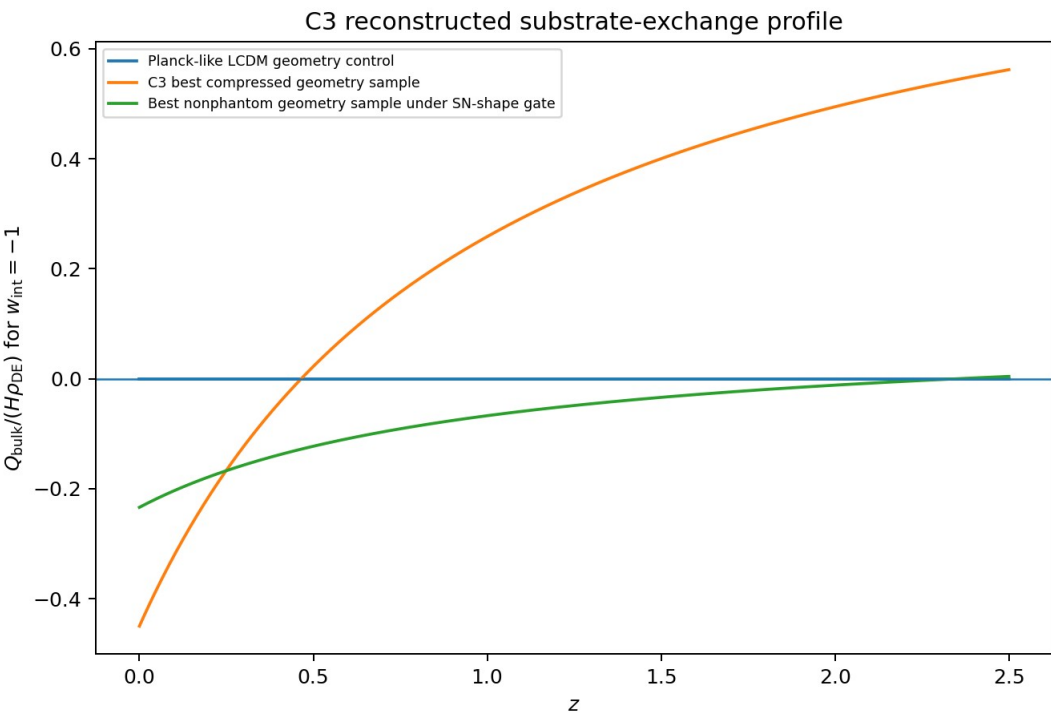


Figure. Selected Q_bulk profiles from the C3 diagnostic scaffold.

9. Falsifiability

- Future tensor limits push well below $r \sim 1e-3$ without detection and without a viable parameter island.
- Preferred n_s exits the BIH-VI island after nuisance and reheating freedom are included.
- A large or opposite-sign α_s is detected.
- Small-scale probes rule out the predicted localized suppression/ringing region.
- Full likelihoods reject $\Delta M_B \sim 0.18$ mag after independent anchors are included.
- $Q_{\text{bulk}}(z)$ reconstruction requires discontinuities, oscillations, or sign-flip chatter.

10. Conclusion

The resulting hierarchy is deliberate: frozen engine, metric-lapse stress test, calibration-frame extension, and Q_{bulk} falsifier. BIH-VI should therefore be understood as an EFT-level cosmological framework with a frozen early-universe branch and a late-universe validation pipeline. It is not presented as a final microscopic substrate theory, nor as a completed solution to the Hubble tension.

Appendix A. Equation map

Geometric seam: $\chi = h^{\{ab\}} \nabla_a R \nabla_b R = 1 - 2G E_{\text{MS}}/R$

Sqrt-Reg: $S_A = 1/[4G \sqrt{1-A_c/A}]$, $F(x)=2x_c[1-\sqrt{1-x/x_c}]$

EFT clock: $H^2=H_c^2 y(2-y)$, $y_N=-m(y)y$, $\epsilon_H=m(1-y)/(2-y)$

C3: $\nu_{\text{metric}}=1$, $H_0_{\text{ladder}}=\nu_{\text{cal}}$ H_0_{geom} , $\Delta M_B=5\log_{10}(\nu_{\text{cal}})$

Appendix B. Rejected branches

- Log-critical entropy: caps H but not density; demoted.
- Pure $f(T)$: scalar/tetrad antisymmetric-constraint pathology.
- Minimal $f(T,B)$: scalar IR recovery failure.
- Analytic $f(R)$: wrong-sign scalaron in analytic IR branch.
- Metric lapse $C1/C2$: SN/BAO distance-gate failure.

Appendix C. Likelihood implementation

Insert repository link, code version, data requirements, convergence criteria, and production posterior tables after running the full C3 likelihood.

Appendix D. Repository structure

See `supplemental/repository_layout.md` in the package.

References

Penrose (1965); Hayward (1994, 1998); Cai and Kim (2005); Bousso and Engelhardt (2015); Cheung et al. (2008); Maldacena (2003); Planck 2018 constraints on inflation; BICEP/Keck constraints; Pantheon+; DESI DR2; Riess/SH0ES; entropy-corrected Friedmann literature; horizon-decoherence literature. Verify all metadata before submission.